

VIRAJ THAKUR

Python Developer | Backend · APIs · GenAI · Docker

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PROFESSIONAL SUMMARY

Python Developer with hands-on experience building production-ready backend systems, REST APIs, async pipelines, and GenAI integrations. Shipped LieLens — a full SaaS platform (Django · Celery · Redis · NLP) — and a FastAPI Finance backend with RBAC and test coverage. Proficient in LangChain, LangGraph, PostgreSQL, Docker, and JWT-based authentication. Practises vibe coding to prototype and ship fast.

TECHNICAL SKILLS

Languages	Python (primary) · SQL · JavaScript
Backend	Django · FastAPI · Flask · REST API · SQLAlchemy · JWT · WebSockets (Socket.IO)
GenAI / AI	LangChain · LangGraph · AI Agents · RAG · Prompt Engineering · NLP · Scikit-learn
Databases	PostgreSQL · MySQL · Redis (Pub/Sub · Caching) · SQLite
DevOps	Docker · Docker Compose · AWS (EC2, S3) · GitHub Actions (CI/CD) · Git
Concepts	OOP · CRUD · RBAC · Async Processing · SaaS Architecture · Microservices · Vibe Coding

PROJECTS

LieLens AI — Writing Risk Intelligence SaaS

[Github/LieLens](#)

- Engineered a full-stack AI SaaS platform scoring writing credibility across 5 NLP+ML dimensions (confidence, exaggeration, emotional intensity, credibility, final risk) with token-level heatmap visualization and downloadable PDF reports.
- Built async processing pipeline using Celery + Redis, cutting analysis latency for heavy NLP tasks; Celery Beat powers automated monthly insight emails and usage resets.
- Implemented SaaS billing scaffold (Stripe webhooks), API key management with plan-based rate limiting, JWT auth, and RBAC — mirrors production SaaS architecture.

Stack: Django · Celery · Redis · PostgreSQL · Scikit-learn · NLP · Docker · Stripe · JWT · GitHub Actions CI

Skillo — Local Skill Marketplace

[Github/Skillo](#)

- Engineered a multi-role marketplace (seeker & provider) with JWT role-based authentication, REST API endpoints, and real-time booking notifications via Socket.IO + Redis pub/sub.
- Containerised full stack (Flask + PostgreSQL + Redis) with Docker Compose for single-command reproducible deployment.

Stack: Flask · PostgreSQL · Redis · JWT · Socket.IO · Docker · REST API

Speech Emotion Recognition

[Github/speechER](#)

- Built end-to-end ML pipeline classifying 6 emotions from raw speech on RAVDESS dataset; achieved 82% accuracy by engineering 40-dim MFCC features and tuning an SVM classifier.
- Evaluated model using Precision, Recall, F1-Score; iterated on preprocessing to improve baseline accuracy by ~12 percentage points.

Stack: Python · Scikit-learn · Librosa · Pandas · NumPy · Matplotlib · Seaborn

EDUCATION

B.E. Computer Engineering

Sindhudurg Shiksha Prasarak Mandal, Kankavli

2021 – 2025

CGPA: 8.20

B.Sc. Mathematics

Balasaheb Khardekar College, Vengurla

2020 – 2021

81.67%

CERTIFICATIONS

Programming in Python — Besant Technologies (Feb 2024)

Generative AI & Model Lifecycle — NXT Wave (Dec 2024)

SQL — DDL / DML / Joins / Aggregations — NXT Wave (Sep 2024)

AWS Solution Architecture Job Simulation — Forage (Oct 2024)

ACHIEVEMENTS

- Shipped LieLens — a full-stack AI SaaS with async architecture, billing scaffold, and explainable AI, built end-to-end as a solo project.
- Completed AWS Solution Architecture Job Simulation (Forage) — designed multi-tier scalable cloud architecture.
- Active contributor — National Digital Library Program (NDLI Club, IIT Kharagpur).